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655.662 — Introduction to Healthcare Systems Engineering
Module 6 | M6A3: Team Learning

How Team Learning Enables the PD-RPM Remote Care System

How can "Team Learning" enable your remote care system?

Senge defines team learning as **the process of aligning and developing the capacity of a team to create the results its members truly desire**, and he is careful that alignment is not agreement. The unaligned team wastes energy: individuals may work extremely hard without that effort becoming team effort, like a light bulb scattering in every direction, where an aligned team is a laser. Alignment is the precondition — empowering individuals in an unaligned team only increases the scatter. He names three dimensions: thinking insightfully about **complex issues**, where many minds must be more intelligent than one; **innovative coordinated action**, which runs on operational trust rather than instruction; and the obligation of a learning team to **propagate the practice to other teams**.

Why PD-RPM needs this more than most programs. Its teams are neither co-located nor co-employed: the home dialysis program, the monitoring-center engineers, the cycler manufacturer's field service, the carrier's network operations, the electronic health record vendor's integration group, and the patients and care partners who operate the system nightly. The project cannot direct most of them — it is an acknowledged system of systems. More importantly, its failures are cross-boundary by construction. The silent-gateway risk carried in the risk register cannot be diagnosed by any single team, because the evidence is split across carrier logs, gateway firmware, the nurse's queue and the patient's kitchen. No one holds enough of it alone.

Seven practices that make it operational. Team learning is a discipline, not a disposition, so it has to be built into how the program meets.

Collaborative practice	Chapter 11 principle it implements	Who takes part	What it produces
Weekly silent-patient review	Dialogue: suspend assumptions — no cause may be named in the first ten minutes	Nurse, firmware lead, carrier engineer	A shared account of what happened, before anyone defends a position
Rank left at the door	Dialogue: regard one another as colleagues	All participants	The nurse's observation weighs the same as the architect's
A named facilitator who does not decide	Dialogue: someone must hold the context	Systems engineer	The conversation stays exploratory long enough to be useful
Converge-and-assign close	Discussion: the team moves deliberately to decision	Same group	One change, one owner, one date
Shared timestamped evidence log	Surfaces defensive routines by removing ownership of the facts	All parties, including the carrier	"Whose fault" becomes "what happened"
Patients and care partners seated in the review	Dimension one: many minds more intelligent than one	Patient, care partner	Access to the half of the system no instrument observes
Fleet replay sandbox	The practice field — a microworld to rehearse in	Engineering and nursing together	Rehearsal of a failure without patient consequence

Dialogue, discussion, and the routines that block both. Senge separates two ways a team converses. *Discussion* shares a root with percussion and concussion: ideas are batted back and forth until one wins, which is how a team converges. *Dialogue* is a free flow of meaning in which people become observers of their own thinking, and Bohm's three conditions — suspend assumptions, treat one another as colleagues, and have someone hold the context — are what the first six practices above implement. A learning team masters the movement between the two. What prevents it is what Argyris calls **defensive routines**, the habitual moves that protect us from embarrassment and seal in mediocrity. In this program the routine is predictable and it is attribution: the carrier says gateway, the firmware lead says carrier, the nurse says the system was down again. Each is defending against being the cause, and the shared evidence log exists to make that defense unnecessary.

Scenario: the phantom Sunday outage. During the clinical pilot, eleven patients logged device-silent events clustered on Sunday nights across three weeks. The carrier found no outage in its logs. Firmware telemetry showed no faults and a clean battery history. The nurse reported that patients said everything was fine. Each team, examining its own evidence honestly, concluded the problem belonged to someone else, and the program came close to writing it off as instrumentation noise — the textbook defensive outcome, where three correct analyses sum to no learning. The first dialogue session seated all three teams plus two pilot patients, with the no-cause-for-ten-minutes rule in force. Nine minutes in, a patient mentioned switching off the power strip before bed on Sundays, after the cleaner came. The gateway battery holds six hours; Sunday sessions run long. No participant could have found this alone: the carrier had no outage, the firmware had no fault, the nurse had no complaint, and the patient did not know the power strip was part of the system. The outcome was a 24-hour battery, a line in the patient instructions, and a standing decision to seat patients in every review.

That last outcome is Senge's third dimension doing its work: the home program has since borrowed the same format for its technique-failure reviews, so the practice is propagating beyond the team that built it. The reason this belongs in advanced development rather than after fielding is that the phase is the first time these groups have to produce anything together. A team that has not learned to talk before integration will not learn during it.